

A Comprehensive Survey on Deep Clustering: Taxonomy, Challenges, and Future Directions

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Clustering is a fundamental machine learning task, which aim at assigning instances into groups so that similar samples belong to the same cluster while dissimilar samples belong to different clusters. Shallow clustering methods usually assume that data are collected and expressed as feature vectors within which clustering is performed. However, clustering high-dimensional data, such as images, texts, videos, and graphs, poses significant challenges for clustering tasks, such as indiscriminate representation and intricate relationships among instances. Over the past decades, deep learning has achieved remarkable success in effective representation learning and modeling complex relationships. Motivated by these advancements, **Deep Clustering** seeks to improve clustering outcomes through deep learning techniques, garnering considerable interest from both academia and industry. Despite many contributions to this vibrant area of research, the lack of systematic analysis and a comprehensive taxonomy has hindered progress in this field. In this survey, we first explore how deep learning can be integrated into deep clustering and identify two fundamental components: the representation learning module and the clustering module. Then, we summarize and analyze the representative design of these two modules. Furthermore, we introduce a novel

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taxonomy of deep clustering based on how these two modules interact, specifically through multistage, generative, iterative, and simultaneous approaches. In addition, we present well-known benchmark datasets, evaluation metrics, and open-source tools to clearly demonstrate different experimental approaches. Finally, we examine the practical applications of deep clustering and propose challenging areas for future research.

CCS Concepts: • **Theory of computation** → **Unsupervised learning and clustering**;

Additional Key Words and Phrases: Deep learning, clustering, representation learning

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1 Introduction

Clustering is a fundamental technique in machine learning, often used as a crucial pre-processing step in various data mining tasks. The primary purpose of clustering is to assign the instances into groups so that similar samples belong to the same cluster while dissimilar samples belong to different clusters. The clusters of samples provide a global characterization of data instances, which can significantly benefit further analysis of the whole dataset, such as anomaly detection [1, 2], domain adaptation [3, 4], community detection [5, 6], and discriminative representation learning [7–9], and so on.

Early clustering works typically assume that data is represented in the form of feature vectors and primarily focus on discovering clustering structures within the vector spaces, which is also known as **shallow clustering**. With the rapid development of the Internet and Web services, the research community is showing increasing interest in clustering more complex data such as text, images, videos, and graphs. However, the raw high-dimensional features of these data pose significant challenges for shallow clustering due to the “curse of dimensionality” [10]. Meanwhile, the past decades have also witnessed the success of deep learning in learning representation for high-dimensional data, especially unsupervised ones. Recently, **Deep Clustering**, which aims to enhance the clustering task through deep learning, has attracted significant interest from both the research community and the industry.

To accomplish this objective, a simplistic approach might involve using deep unsupervised representation learning to learn representations for each data instance, and subsequently input these representations into shallow clustering methods to achieve the ultimate clustering results. However, this straightforward method fails to recognize the fundamental link between representation learning and clustering: clustering offers crucial insights for representation learning, while discriminative representation significantly improves the clustering procedure. Given these motivations, deep clustering approaches are dedicated to addressing these research questions:

- (1) How to learn discriminative representations with deep learning that can yield better clustering performance?
- (2) How to acquire accurate clustering results given discriminative representations with deep learning?
- (3) How to efficiently conduct clustering and representation learning in a unified framework and enhance each other?

In light of the three challenges mentioned above, numerous deep clustering methods have emerged, incorporating diverse deep architectures and data types, and have achieved significant

Table 1. Comparison with Related Surveys

References Comparisons		Shallow Clustering					Deep Clustering						
		[13]	[19]	[14]	[12]	[15]	[16]	[17]	[18]	[20]	[21]	[22]	Ours
Deep representation learning design	Auto Encoder	–	–	–	–	–	✓	✓	✓	✓	✓	✓	✓
	Generative	–	–	–	–	–	✓	✓	–	✓	✓	✓	✓
	Contrastive	–	–	–	–	–	–	–	–	–	✓	✓	✓
	Cluster Based	–	–	–	–	–	✓	✓	–	–	✓	–	✓
Clustering with DNN		✓	✓	✓	–	✓	✓	✓	✓	✓	✓	✓	✓
Interaction between Representation and Clustering	Multistage	✓	✓	✓	✓	✓	–	✓	✓	✓	✓	✓	✓
	Iterative	✓	✓	–	–	–	–	✓	✓	–	–	–	✓
	Generative	–	–	–	–	–	✓	–	–	✓	✓	✓	✓
	Simultaneously	–	–	–	–	–	✓	✓	✓	–	✓	✓	✓
Application		✓	✓	✓	–	✓	–	–	–	–	–	✓	✓
Various types of Dataset		–	–	–	–	–	–	–	–	–	–	–	✓
Evaluation		–	–	–	✓	✓	✓	–	–	–	–	✓	✓
Implementation		–	–	–	–	–	–	–	–	–	–	✓	✓

*Each column indicates a survey paper which is being compared, “✓” means the term of the corresponding row has been surveyed or analyzed in this paper while ‘–’ means not. Note that surveys published later than ours are also included.

success. However, the existing literature has not sufficiently synthesized the current research landscape, limiting the formation of a detailed understanding that effectively integrates and assesses the varied results in this area. To address this deficiency, this article presents a comprehensive survey of deep clustering, introducing a novel taxonomy of existing methods, examining unresolved challenges, and suggesting essential directions for future research.

Related Surveys. As a classic task, clustering has experienced extensive development in the past decades, with numerous surveys focusing on shallow clustering [11–15]. Typically, shallow clustering methods discussed in these surveys utilize feature representations as input and produce cluster assignments for each data instance. Subsequently, as deep learning has evolved, several studies have begun to explore clustering techniques that incorporate deep learning approaches.

Min et al. [16] state that the essence of deep clustering is to learn clustering-oriented representations, so the literature should be classified according to the network architecture of representation learning. Aljalbout et al. [17] designed the taxonomy mainly based on the representation learning architecture and the loss function, discussing the comparisons between methods only on the MNIST and COIL20 datasets. Nutakki et al. [18] review some clustering methods with deep learning, while the most recent advanced techniques for representation learning and clustering were excluded. Furthermore, all these works [16–18] merely cover outdated methods and miss more recent advances in this active research area.

Nevertheless, previous surveys fail to provide a thorough analysis of how deep learning is integrated into deep clustering approaches, or to offer a detailed classification of the methods involved. This deficiency has impeded the advancement of deep clustering and urgently requires rectification. In this article, we focus on clustering techniques that utilize deep learning, particularly focusing on how deep representation learning interacts with clustering using deep neural networks. A summary of how our survey compares to the previous ones is presented in Table 1. Note that some references were cited after the initial pre-print date of this article.

Contributions. In summary, this article seeks to offer a comprehensive survey of the latest deep clustering techniques and help potential readers understand the broad landscape of deep clustering with respect to the following aspects:

- **Cornerstones of Deep Clustering.** We summarize two cornerstones of deep clustering, namely the representation learning module and the clustering module. For each module, we highlight the representative and universal designs summarized from existing methods, which can be easily generalized to new models.
- **Systematic Taxonomy.** We propose a systematic taxonomy of existing deep clustering methods based on the ways of interactions between the representation learning module and the clustering module providing four representative branches of methods. We also compare and analyze the properties of each branch in different scenarios.
- **Abundant Resources and References.** We collect various types of benchmark datasets, evaluation metrics, and open-source implementations of the latest publications on deep clustering, which are organized together with references on Github.¹ We also provide an open source and easy-to-use package that can effectively deploy and develop deep clustering methods, with the aim of further advancing this active research area.²
- **Future Directions.** Based on the properties of the representation learning module and the clustering module, as well as their interactions, we discuss the limitations and challenges of existing methods, followed by our insights and thoughts on promising research directions that deserve future investigations.

Organization. The rest of this survey is organized as follows: Section 2 introduces the basic definitions and notation used in this article. Section 3 summarizes the representative design of the representation module, along with different data types. Section 4 summarizes the representative design of the clustering module, which focuses mainly on the basic modules defined in the deep clustering methods. Section 5 summarizes the representative ways of interactions between the two modules, which cover most of the existing literature. Section 6 introduces the widely used benchmark datasets and evaluation metrics. Section 7 discusses the applications of deep clustering. Section 8 discusses limitations, challenges and suggests future research directions that deserve further exploration. The overall organization of this survey is illustrated in Figure 1.

2 Preliminary

In this section, we first briefly introduce some definitions in deep clustering that need to be clarified, then illustrate the notation used in this article in Table 2.

Deep Clustering and Shallow Clustering. Given a set of data instances $X = \{x_i\}_{i=1}^N$, the objective of clustering is to automatically assign each instance x to groups so that instances in the same group are similar while instances from different groups are dissimilar. The *shallow (non-deep) clustering* takes the feature vectors of the instances as input and outputs the clustering result without deep neural networks. The *deep clustering* aims to cluster unstructured data or high-dimensional data with deep neural networks. It is worth noting that deep clustering is not narrowly defined as applying deep learning techniques in representation learning. Instead, clustering itself can be conducted by deep neural networks and benefits from the interaction with deep representation learning.

Hard Clustering and Soft Clustering. The clustering methods can be categorized into hard and soft clustering according to the type of output they generate. The output of hard clustering is a discrete one-hot cluster label $\tilde{y}_i$ for each instance x_i , while the output of soft clustering is a

¹<https://github.com/zhoushengisnoob/DeepClustering>

²<https://github.com/zhoushengisnoob/OpenDeepClustering>

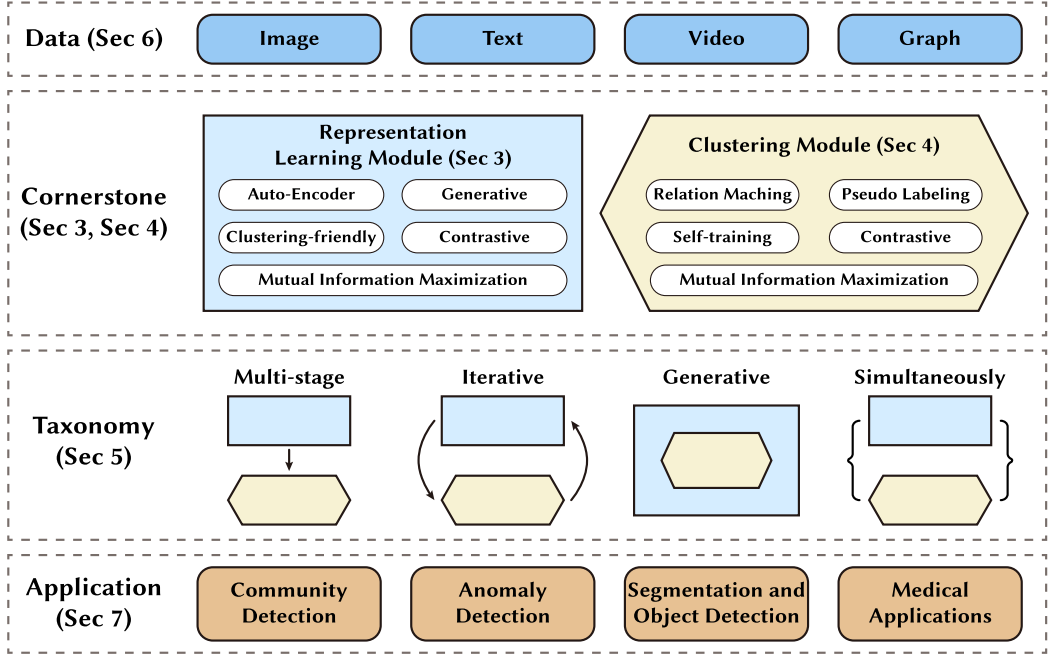


Fig. 1. Overall organization of this survey. We use blue and yellow to highlight the representation of learning module and clustering module.

Table 2. Important Notations Used in this Article

Notations	Explanations	Notations	Explanations
N	The number of data instances	K	The number of clusters
x	Data instance	$\hat{x}$	Data reconstruction
$x^{\mathcal{T}}$	Augmented instance	$\cdot^{\mathcal{T}}$	Transpose of a matrix (vector)
h	Representation	z	Soft assignment
$\tilde{y}$	Predicted hard label	y	Ground truth label
c	Representation (centroid) of cluster	τ	Temperature parameter
$\mathbb{P}, \mathbb{Q}$	Probability distribution	$\mathcal{X} = \{x_i\}_{i=1}^N$	Data instances set
$\ \cdot\ _F$	Frobenius norm of a matrix (vector)	$ \cdot $	The number of elements in a set

continuous cluster assignment probabilistic vector $z_i \in \mathcal{R}^K$. The primary objective of clustering is to assign samples into distinct groups, hence the outcomes of clustering are typically presented with discrete labels. However, the discrete assignment of an instance is usually hard to optimize, especially for deep neural networks with backpropagation. As a result, most of the existing deep clustering methods fall into the soft clustering category. In this approach, the clustering results are produced by a deep neural network f , which produces logits of K dimensional activated by a softmax function. It should be noted that the soft clustering results can be easily transformed to hard clustering by selecting the cluster with the highest probability.

Partitioning Clustering and Overlapping Clustering. The previously described hard clustering, which maximizes the probability vector, is intended for the *partitioning clustering*

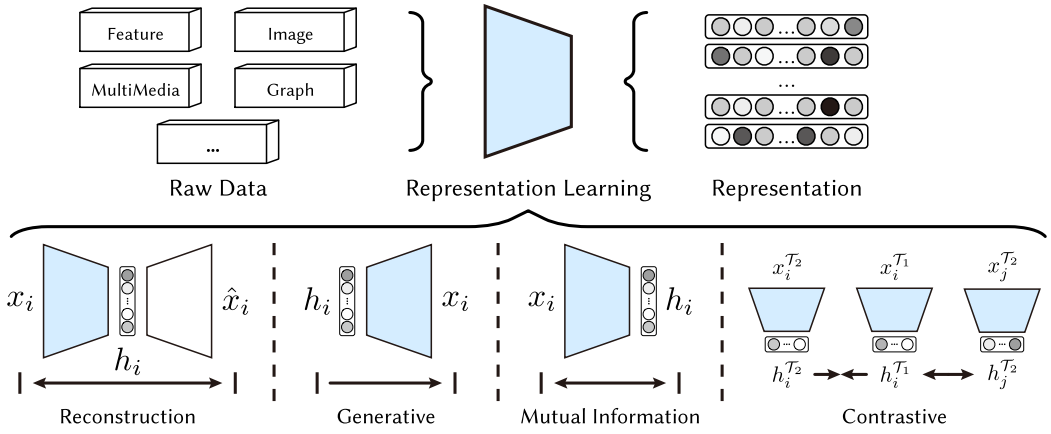


Fig. 2. Representative representation learning modules that transform raw data into lower-dimensional representations.

scenario, where each data instance is assigned to a single cluster only. Consequently, this clustering approach divides the entire dataset into K separate groups. Most current deep clustering techniques are non-overlapping, reflecting the nature of the primarily non-overlapping datasets they are tested on. In the context of overlapping clustering, a data instance can be part of multiple clusters, introducing additional challenges to clustering methods, which will be explored further in Section 8.

3 Representation Learning Module

The past few decades have seen significant advancements in deep representation learning [23–25], particularly in unsupervised methods. All unsupervised learning approaches can function as input generators and are seamlessly integrated into the deep clustering framework, with more details provided in Section 4. However, these methods are generally not designed for clustering tasks and do not incorporate essential clustering data to enhance representation learning. This section presents the *representation learning module* used in deep clustering, which processes raw data to produce a compact, low-dimensional representation. Figure 2 shows the key representation learning modules discussed in this section.

3.1 Auto-Encoder Based Representation Learning

Auto-Encoder [26] is one of the most widely adopted unsupervised deep representation learning methods due to its simplicity and effectiveness. The auto-encoder is a linear stack of two deep neural networks named encoder and decoder. The encoder f_e encodes the input data x in a low-dimensional representation $h = f_e(x)$, and the decoder f_d decodes the low-dimensional representation h to the input data space $\hat{x} = f_d(h)$. A good auto-encoder is expected to reconstruct the input data without dropping any information, thus the optimization target of the auto-encoder is usually formulated as Equation (1):

$$\mathcal{L}_{AE} = \sum_i^N \|x_i - \hat{x}_i\|_2^2 = \sum_i^N \|x_i - f_d(f_e(x))\|_2^2, \quad (1)$$

where $\|\cdot\|_2$ is the L2-norm, $\hat{x}_i$ is the reconstructed data. The auto-encoder is a general structure and can be customized for different data types. For example, deep neural networks for vectorized

features [27], convolutional networks for images and graph neural networks for graphs [28], 3D convolutional networks and LSTM auto-encoder for videos [29] and so on.

Analysis. The representation learning framework based on auto-encoders is appreciated for its straightforward implementation and effective training, and has been extensively used in initial deep clustering studies. However, this approach learns representations on an individual instance basis, often overlooking the interconnections between different instances. Consequently, in the embedding space, instances may not be distinctly separated, leading to suboptimal clustering results.

3.2 Deep Generative Representation Learning

Another line of deep unsupervised representation learning lies in generative methods which assume that the data x are generated from a latent representation h and then reversely deduce the posterior of the representation $p(h|x)$ from the data. Among these, the most typical method is the **Variational Auto-Encoder (VAE)** [30] and **Generative Adversarial Networks (GAN)** [31]. Both methods focus on capturing the data distribution, which is used to map the input data into the latent embedding space. VAE resorts to the variance inference technique and maximizes the **evidence lower bound (ELBO)** on the data likelihood:

$$\log p(x) \geq \mathbb{E}_{q(h|x)}[\log p(x|h)] - D_{KL}(q(h|x)||p(h)), \quad (2)$$

where $D_{KL}(\cdot||\cdot)$ denotes the KL-divergence between two distributions, $p(h)$ is the prior distribution of the latent representation, $q(h|x; \varphi)$ is the variational posterior of the representation to approximate the true posterior (i.e., $q(h|x; \varphi) \approx p(h|x)$), which can be modeled with the recognition networks φ . Using the reparameterization trick [30] and the Monte Carlo approximation [32], the posterior can be efficiently learned from the Equation (2) via backpropagation. GANs learn representations via a competitive mechanism in which a generator network generates samples to deceive a discriminator network. In response, this discriminator network learns to differentiate between real and generated data, forcing the generator to progressively improve its representations. By modeling the data distribution in an adversarial way, the GANs can learn more discriminative representation, thereby enhancing their clustering capabilities.

Analysis. Deep generative representation learning exhibits advantages such as flexibility, interpretability, and the ability to replicate data points. However, in comparison to other methods of representation learning, they struggle to incorporate clustering information effectively during the learning phase. The widely used strategy involves making assumptions at the cluster level about the data distribution, such as using Gaussian mixture models. However, these assumptions often do not suffice for handling complex data distributions. Consequently, exploring ways to embed category information within the distribution modeling phase continues to be an essential question for deep generative representation learning in clustering contexts.

3.3 Mutual Information Maximization Representation Learning

Mutual information (MI) [33] is a fundamental quantity to measure the dependence between random variables X and Y , which is formulated as Equation (3):

$$I(X; Y) = \int \log \frac{d\mathbb{P}_{XY}}{d\mathbb{P}_X \otimes \mathbb{P}_Y} d\mathbb{P}_{XY}, \quad (3)$$

where $\mathbb{P}_{XY}$ is the joint distribution, $\mathbb{P}_X = \int_Y d\mathbb{P}_{XY}$ and $\mathbb{P}_Y = \int_X d\mathbb{P}_{XY}$ are the marginal distribution, $\mathbb{P}_X \otimes \mathbb{P}_Y$ is the product of the marginal distributions. Traditional mutual information estimations [34] are only tractable for discrete variables or known probability distributions. Recently, MINE [35] was proposed to estimate mutual information with deep neural networks. The widely

used mutual information estimation is the **Jensen-Shannon divergence (JSD)** [36] formulated as Equation (4):

$$\mathcal{I}_{JSD}(X; H) = \mathbb{E}_{\mathbb{P}_{XH}}[-\text{sp}(-D(x, h))] - \mathbb{E}_{\mathbb{P}_X \times \mathbb{P}_H}[\text{sp}(D(x, h))], \quad (4)$$

where $\text{sp}(x) = \log(1 + e^x)$ is the softplus function. D is a discriminator function modeled by a neural network. Another popular mutual information estimation is InfoNCE [37], which will be introduced in Section 3.4. Taking advantage of neural estimation, mutual information has been widely adopted in unsupervised representation learning [38, 39]. More specifically, the representation is learned by maximizing mutual information between different layers [39] or different parts of the data instances [38], so that the consistency of the representation can be guaranteed. This can be viewed as an early attempt at self-supervised learning which has an extensive impact on the later work.

Analysis. The significant benefit of this branch of methods is that the variables assessed through mutual information are not confined to identical dimensions or semantic spaces, like instances and clusters. Similar to auto-encoder based and deep generative representation learning, the mutual information maximization approaches are also focused on individual instances, potentially facing the same issues in modeling the relationships among instances. However, the marginal distribution used in mutual information estimation is influenced by all observed samples and the relationships among instances are indirectly established.

3.4 Contrastive Representation Learning

Contrastive learning is one of the most popular unsupervised representation learning techniques in recent years. The basic idea is to pull a positive pair of instances close while pushing a negative pair of instances far away, which is also known as instance discrimination. The representative target of contrastive learning is the InfoNCE loss [37] formulated as Equation (5):

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{|N|} \sum_{i=1}^N \log \frac{\exp(f(h_i^{\mathcal{T}_1}, h_i^{\mathcal{T}_2})/\tau)}{\sum_{j=1}^N \exp(f(h_i^{\mathcal{T}_1}, h_j^{\mathcal{T}_2})/\tau)}, \quad (5)$$

where $h_i^{\mathcal{T}_1}$ and $h_i^{\mathcal{T}_2}$ is a pair of positive samples and $h_j^{\mathcal{T}_2}$ are the negative sample representations, f is a similarity function, τ is the temperature parameter [40]. The positive samples are usually conducted by data augmentation which may vary from different data types. For example, flip, rotate, and crop augmentation for image data [41], node dropping, edge perturbation, attribute masking, and subgraph sampling for graph data [42, 43]. Negative samples are selected from an augmented view of other instances in the dataset [41] or from a momentum-updated memory bank of old negative representations [44], which can be viewed as an approximation of noise.

Analysis. Numerous theoretical studies [45–47] have been conducted on contrastive learning, demonstrating that contrastive learning tends to pull instances with the same label close while pushing instances with different labels away in the low-dimensional embedding space. Consequently, this type of learning can greatly improve the effectiveness of clustering by enhancing discriminative representation. In addition, it has been proved that reducing InfoNCE loss corresponds to an increase in the lower limit of mutual information [37]. Thus, contrastive learning can also be used to establish explicit connections between instances and clusters.

3.5 Clustering Friendly Representation Learning

Although the aforementioned representation learning methods have somehow implicitly boosted the performance of clustering, they are not explicitly designed for the clustering task. In this

subsection, we will introduce some representation learning methods that explicitly support the clustering task.

K-means [48] friendly representation is first defined in DCN [49] where samples in the low-dimensional space are expected to scatter around their corresponding cluster centroids. This can well support the assumption of K-means that each sample is assigned to the cluster with the minimum distance to the centroid. The objective $\mathcal{L}_{KF}$ can be formulated as Equation (6):

$$\mathcal{L}_{KF} = \sum_{i=1}^N \|\mathbf{f}(x_i) - \tilde{y}_i \mathbf{C}\|_2^2, \quad (6)$$

where $\mathbf{f}(\cdot)$ is the deep neural network for representation learning, $\tilde{y}_i$ is the hard assignment vector of data instance x_i which has only one non-zero elements, $\mathbf{C}$ is the cluster representation matrix where k th column of $\mathbf{C}$, i.e., c_k denotes the centroid of the k th cluster.

Spectral clustering friendly representation learning is inspired by the eigen decomposition in spectral clustering [50] that projects instances into a space with orthogonal bases. In deep clustering, the orthogonal basis is modeled explicitly by reducing correlations within features [51], which can be formulated as Equation (7):

$$\mathcal{L}_{SF} = \sum_{m=1}^d \left(-h_m^T h_m / \tau + \log \sum_n \exp(h_n^T h_m / \tau) \right), \quad (7)$$

where d is the number of feature dimensions, h_m is the m th dimension feature vector, τ is the temperature parameter. The objective is to learn the independent features so that redundant information is reduced.

Analysis. The clustering friendly representation learning benefits from direct optimization for clustering, which may significantly improve the corresponding cluster performance. However, such simplicity also limits the generalization to other clustering methods. Currently, the research community has made more efforts to inspire clustering methods and express them from a deep learning perspective, rather than learning specific representations for each clustering method.

3.6 Subspace Representation Learning

Subspace representation learning is the early stage of subspace clustering [52], which aims to map the data instances to a low-dimensional subspace where instances can be separated. Basically, current subspace representation learning methods [53–58] have relied on the self-expressiveness assumption, where a data instance can be expressed as a linear combination of other data instances from the same subspace, i.e., $\mathbf{X} = \mathbf{X}\Theta_c$, where $\mathbf{X}$ is the data matrix and Θ_c is the matrix of self-expression coefficients. For representation learning, the self-expression property leads to the following objective as in Equation (8):

$$\min_{\Theta_c} \|\Theta_c\|_p + \frac{\lambda}{2} \|\mathbf{H} - \mathbf{H}\Theta_c\|_F^2 \quad \text{s.t.} \quad \text{diag}(\Theta_c) = \mathbf{0}, \quad (8)$$

where $\|\cdot\|_p$ is a matrix norm, λ controls the weight balance. $\mathbf{H}$ denotes the sample representations learned by a network. In addition, Θ_c can be implemented as parameters of an additional network layer [55].

Analysis. The effectiveness of subspace representation learning is based on the premise that data can be represented through various subspaces characterized by unique feature sets, capable of managing datasets with diverse dimensions and intricate relationships. However, it is challenged by computational difficulties, such as determining the best subspaces and allocating data points to these subspaces, particularly in the context of extensive, high-dimensional datasets.

3.7 Data Type Specific Representation Learning

The representation learning techniques mentioned above are suitable for various data types, but in practical applications, some methods are designed specifically for certain data types. This subsection offers a summary of four widely studied data types and their corresponding representation learning strategies in the context of deep clustering.

3.7.1 Image Representation Learning. Learning representations of images using CNN [59] and ResNet [60] as a backbone has achieved great success in the past decades. In the image deep clustering, they still play active roles as feature extractors or backbones in the representation learning module. Beyond the above two methods, recent advances have been made by introducing modern representation learning techniques such as vision transformer [61] to deep clustering. As one of the most popular directions, unsupervised representation learning for image data will play a central role in deep clustering and affect the other data types.

3.7.2 Text Representation Learning. The early attempts of text representation learning have utilized statistical-based methods such as TF-IDF [62], Word2Vec [63], and Skip-Gram [64]. Later, some works focus on topic modeling [65] and semantic distances [66, 67] for text representation learning, and more [68] on unsupervised scenarios. Recently, pre-trained language models like BERT [69] and GPT-3 [70] are gradually dominating the area of text representation learning. However, the fine-tuning [71] of these methods in the deep clustering task remains an open question. Furthermore, while **large language models (LLMs)** [72] have shown remarkable linguistic prowess in various NLP applications, their utilization is restricted to APIs. The use of LLMs for sideway information for clustering has shown great potential [73].

3.7.3 Video Representation Learning. Video representation learning presents a complex challenge, integrating spatial-temporal learning, multimodal learning involving audio [74], and natural language processing tasks such as video abstracts and subtitles, all within a single framework. The early methods utilize the LSTM Autoencoder [29], 3D-ResNets [75], and 3D-U-Net [76] as feature extractor. Recent methods have focused on spatial-temporal modeling [29, 77–80] and Qian et al. [75] in particular incorporates contrastive learning for self-supervision.

3.7.4 Graph Representation Learning. The classic graph representation learning aim at learning low-dimensional representation for nodes so that the proximity among nodes can be preserved in the embedding space. **Graph Neural Networks (GNNs)** [81, 82] are widely used, including GCN [83], GraphSAGE [84] and GAT [85], and it gives infinite possibilities of graph node representation learning combining node features and graph topology [86–89]. Furthermore, graph-level information also has great potential in tasks such as protein classification [90], which has prompted increasing attention to graph-level representation learning [91–94].

Analysis. The data type specific representation learning mentioned above can be a naive backbone for feature extraction or end-to-end unsupervised representation learning, which are the most active research directions in deep learning. With more types of data being collected and the fast development of deep learning, we believe that deep clustering will grow along with the data type-specific representation learning techniques.

4 Clustering Module

In this section, we present the key clustering modules used in deep clustering, which utilize low-dimensional representations as input, and produce either cluster labels for hard clustering or probabilities of cluster assignments for soft clustering. Although numerous shallow clustering techniques can be applied directly to clustering tasks, integrating them with deep representation

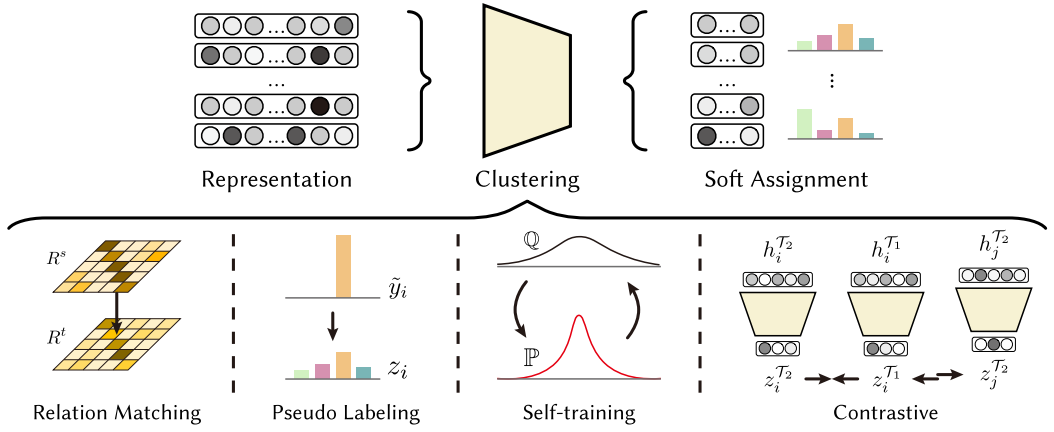


Fig. 3. Representative deep clustering modules that produce soft assignments for samples based on their representations.

learning within a single framework is challenging. Furthermore, these methods often fail to effectively engage the representation learning module to achieve mutual improvement. For further information on shallow clustering methods, we suggest previous surveys [13, 19]. In deep clustering, the objective is to adhere to a deep learning framework, where the clustering results are obtained by optimizing the neural network. As depicted in Figure 3, the subsequent sections will discuss five prevalent approaches to constructing clustering modules in deep clustering.

4.1 Relation Matching Deep Clustering

Relation matching techniques are those that enhance clustering outcomes by aligning connections among instances (instance to instance, I2I) with respect to the clustering outcomes (instance to cluster centers, I2C). To obtain the relationship between instance and cluster centers, the direct approach is to measure the distance or similarity between instance and cluster centers in the low-dimensional space. A closer relation implies a higher probability of belonging to a particular cluster. The I2C can be directly optimized by

$$\mathcal{L}_{I2C} = \sum_{i=1}^N \|h_i - Ms_i\|_2, \text{ s.t. } 1^T s_i = 1 \quad \forall i, j, \quad (9)$$

where s_j is the assignment vector, and the k th column of M , i.e., m_k , denotes the centroid of the k th cluster, h_i is the representation of instance in the low-dimensional space. By fixing x and M , optimizing Equation (9) outputs the cluster assignment; by only fixing x , optimizing Equation (9) is equal to the result of running K-means.

Also, the clustering outcomes can be optimized by matching relationships among instances from different views, which can be formulated as

$$\mathcal{L}_{RM} = \sum_i^N \sum_j^N \ell(R_{ij}^s, R_{ij}^t), \quad (10)$$

where ℓ is a measure of relation matching, e.g., cosine similarity or Euclidean distance, R^s and R^t are the relations in source and target space. Here, the relationship based on I2C is typically treated as the source space, and the target space may come from the similarity between the embedding or raw features of instances.

Analysis. The relation matching deep clustering explicitly connects the representation learning and clustering by matching the relation from embedding spaces to label spaces, which is straightforward and easy to implement. However, calculating N^2 pairs of instances for I2I formulations is computationally inefficient. To tackle this challenge, some methods only preserve the k-nearest-neighbor relations [95, 96] for each instance or the relations with high confidence [95]. Although this can somehow improve the training efficiency, the extra hyper-parameter is hard to set in an unsupervised manner. Furthermore, among all pairs of relations, many of them are noisy especially in the early training phases with limited capability. How to filter out the clean relations to boost the performance while discarding the noisy relations is still an open research question.

4.2 Pseudo Labeling Deep Clustering

Pseudo labeling, a common approach in semi-supervised learning [97], has been recently applied to deep clustering. This method can be considered a form of relation matching, characterized by discrete relations that depend on the consistency of the labels. According to the way of using pseudo labels, existing methods can be largely divided into two groups: instance-wise pseudo labeling [95, 98, 99] and relation-wise pseudo labeling [100, 101].

The instance-wise pseudo labeling filters out a subset of instances with high confidence and trains the network in a supervised manner with cross-entropy loss as Equation (11):

$$\mathcal{L}_{IPL} = -\frac{1}{|\mathcal{X}^c|} \sum_i \sum_{k=1}^K \tilde{y}_{ik} \log(z_{ik}), \quad (11)$$

where $\mathcal{L}_{IPL}$ denotes the loss of instance-wise pseudo labeling, $\mathcal{X}^c$ denotes the subset of instances with high confidence, $\tilde{y}_{ik}$ and z_{ik} are the predicted hard label and soft cluster assignment. The confidence is usually estimated by the entropy or maximum of the probabilistic distribution of assignment.

The general idea of relation-wise pseudo labeling is to enforce instances with same pseudo labels closer while instances with different pseudo labels away from each other in the embedding space. Given the filtered instances, relation-wise pseudo labeling constructs the discrete relations among instances to guide the representation learning: the *must-link* for pairs of instances with same pseudo labels and *cannot-link* for pairs of instances with different pseudo labels as Equation (12):

$$\mathcal{L}_{RPL} = \frac{1}{|C|} \sum_{\{i,k\} \in C} R_{ik} - \frac{1}{|\mathcal{M}|} \sum_{\{i,j\} \in \mathcal{M}} R_{ij}, \quad (12)$$

where $\mathcal{M}$ is the set of must-link relations and C is the set of cannot-link relations, R_{ij} is the similarity of instance x_i and x_j in the low-dimensional embedding space.

Analysis. Pseudo labeling introduces the advantages of semi-supervised learning to unsupervised clustering tasks. However, its effectiveness largely depends on the quality of the pseudo labels, which are prone to the influence of model performance and the adjustment of hyperparameters, particularly in unsupervised settings. Existing approaches [95, 98] employ pretraining as an initial step prior to pseudo labeling to address these issues, yet this area still requires further focus.

4.3 Self-Training Deep Clustering

The self-training strategy is introduced to the deep clustering task [102] and opens up an active branch of methods named self-training deep clustering [9, 102–107]. More specifically, the cluster assignment distribution is optimized by minimizing the KL-divergence with an auxiliary

distribution as Equation (13):

$$\mathcal{L} = D_{KL}(\mathbb{P} \parallel \mathbb{Q}) = \sum_i^N \sum_k^K p_{ik} \log \frac{p_{ik}}{q_{ik}}, \quad (13)$$

where $\mathbb{Q}$ is the cluster assignment distribution and $\mathbb{P}$ is the auxiliary distribution. q_{ik} and p_{ik} denote the probability that instance x_i belongs to the group k . The assignment distribution $\mathbb{Q}$ follows the assumption of K-means and is produced by the embedding distance between the instance and cluster centroids as in Equation (14):

$$q_{ik} = \frac{(1 + \|h_i - c_k\|_2^2 / \alpha)^{-\frac{\alpha+1}{2}}}{\sum_j^K (1 + \|h_i - c_j\|_2^2 / \alpha)^{-\frac{\alpha+1}{2}}}, \quad (14)$$

where h_i is the representation of data instance x_i and c_k is the representation of cluster k , α is the freedom degree of the Student's t-distribution [108]. The auxiliary distribution $\mathbb{P}$ is a variant of the assignment distribution $\mathbb{Q}$ with both instance-wise and cluster-wise normalization as in Equation (15):

$$p_{ik} = \frac{q_{ik}^2 / f_k}{\sum_j^K q_{ij}^2 / f_j}, \quad (15)$$

where $f_k = \sum_i^N q_{ik}$ are soft cluster frequencies.

Analysis. The success of self-training deep clustering relies on the following properties: First, the square of cluster assignment probability with cluster-wise normalization will encourage the model to pay more attention (gradient) to the instances with higher confidence, which in turn reduces the impact of low-confidence ones. As a result, the cluster assignment vector tends to be one-hot. Second, the soft cluster frequencies f_k can be viewed as the sum of the probability that the instance belongs to the k th cluster. This can prevent the degenerate solution that all instances belong to the same cluster. Despite its effectiveness, this objective is prone to class imbalance issues and requires additional investigation.

4.4 Contrastive Deep Clustering

Similar to contrastive representation learning, the target of contrastive deep clustering is to pull the positive pairs close while pushing the negative pairs far away. The major difference lies in the definition of positive pairs and negative pairs, which can be further divided into three groups:

4.4.1 Instance-Instance Contrast. The instance-instance contrast treats the cluster assignment of each instance as the representation and directly reuses the contrastive representation learning loss as Equation (16):

$$\mathcal{L}_{IIC} = -\frac{1}{|N|} \sum_{i=1}^N \log \frac{\exp(f(z_i^{\mathcal{T}_1}, z_i^{\mathcal{T}_2})/\tau)}{\sum_{j=1}^N \exp(f(z_i^{\mathcal{T}_1}, z_j^{\mathcal{T}_2})/\tau)}, \quad (16)$$

where $z_i^{\mathcal{T}_1}$ is the cluster assignment of augmented instance $x_i^{\mathcal{T}_1}$ predicted by the clustering module.

4.4.2 Cluster-Cluster Contrast. The cluster-cluster contrast treats each cluster as an instance in the embedding space, the target is pulling the cluster and its augmented version close while pushing different clusters far away, which can be formulated as Equation (17):

$$\mathcal{L}_{CCC} = -\frac{1}{|N|} \sum_{k=1}^K \log \frac{\exp(f(c_k^{\mathcal{T}_1}, c_k^{\mathcal{T}_2})/\tau)}{\sum_{j=1}^K \exp(f(c_k^{\mathcal{T}_1}, c_j^{\mathcal{T}_2})/\tau)}, \quad (17)$$

where $c_k^{\mathcal{T}_1}$ and $c_k^{\mathcal{T}_2}$ are the representations of cluster k in different augmented views. It should be noted that the cluster-cluster contrast satisfies the basic requirement of clustering that each cluster should be dissimilar, which is in agreement with the clustering-friendly representation learning described in Section 3.5.

4.4.3 Instance-Cluster Contrast. The instance-cluster contrast is similar to the K-means, which utilizes the cluster centroid as explicit guidance. Given the representation of each instance and cluster centroid in the same low-dimensional space, each instance is expected to be close to the corresponding cluster centroid while far from the other cluster centroids. Such similarity and dissimilarity can be naturally modeled by the contrastive learning as Equation (18):

$$\mathcal{L}_{ICC} = -\frac{1}{|N|} \sum_{i=1}^N \log \frac{\exp(f(h_i, c'_i) / \tau)}{\sum_{j=1}^K \exp(f(h_i, c_j) / \tau)}, \quad (18)$$

where c'_i is the corresponding cluster centroid of x_i which is usually estimated by an alternative clustering method. This can also be understood as maximizing the mutual information between the representation and cluster assignment with data augmentations.

Analysis. Besides the advantages inherited from mutual information maximization clustering, the primary advantages of contrastive deep clustering are that data augmentation helps improve the robustness of clustering with data augmentation, which has been ignored by most existing methods. For detailed advantages of contrastive learning, we suggest the former contrastive learning surveys [109, 110].

5 Taxonomy of Deep Clustering

In this section, we expand upon the core modules of representation learning and clustering as outlined in previous discussions on deep clustering frameworks. Given the inputs and outputs of these modules, various strategies are implemented in existing deep clustering methods to enhance their interaction and collaborative efforts, thus increasing the effectiveness of deep clustering. Therefore, we will omit the intricate details of each module's design and instead concentrate on their interconnections, classifying the existing approaches into four specific categories:

- (1) Multi-stage deep clustering: the representation learning module is sequentially connected with the clustering module.
- (2) Iterative deep clustering: the representation learning module and the clustering module are iteratively updated.
- (3) Generative deep clustering: the clustering module is modeled as a prior of representation module.
- (4) Simultaneous deep clustering: the representation learning module and the clustering module are updated simultaneously.

We select the most representative methods from each category within our taxonomy and provide a detailed presentation of their specific designs under our taxonomy in Table 3.

5.1 Multistage Deep Clustering

Given the modules for representation learning and clustering, it is evident that the output of representation learning smoothly transitions into the input of the clustering module, thus directly helping to produce the clustering results. Multistage deep clustering refers to the methods in which the two modules are optimized separately and connected sequentially. With the evolving of both representation learning and clustering modules, multistage methods continue to be an active area in the field of deep clustering.

Table 3. Specific Designs of Representation and Clustering Modules of Methods Following Our Taxonomy

	Representation Module	Clustering Module	Method
Multi Stage	Auto-Encoder	k-means	SAE [111]
	Auto-Encoder	k-means + Relation Matching	DEN [112]
	Subspace	k-means	PARTY [57]
	Contrastive	k-means	IDFD [51]
	Auto-Encoder + Subspace	spectral clustering	DSC-Nets [55], SENet [56]
	Generative + Subspace	spectral clustering	DASC [58]
	Data Specific	Relation Matching	GTL [113]
	Data Specific	spectral clustering	SCPSO [114]
Generative	Data Specific	k-means	ClusterLLM [73]
	Generative (VAE)	Relation Matching	GMVAE [115], VaDE [116]
	Generative (VAE)	k-means + Pseudo Labeling	DGG [117]
	Generative (GAN)	Relation Matching	ClusterGAN ₁ [118], GM-GAN [119], GANMM [120]
	Generative (GAN)	k-means	ClusterGAN ₂ [121]
Iterative	Data Specific + Generative(GAN)	Relation Matching	CommunityGAN [122]
	Pseudo Labeling	Relation Matching	DeepCluster [99], DeepMCAT [123], CCFC [124]
	Auto-Encoder + Clustering Friendly	Relation Matching	DCN [49], AEDC [125]
	Auto-Encoder + Subspace	Relation Matching	NCSC [54]
	Auto-Encoder + Subspace	Pseudo Labeling	S ² ConvSCN [53], PSSC [126]
	Contrastive	Pseudo Labeling + Relation Matching	JULE [7], GATCluster [101]
	Contrastive	Pseudo Labeling	SPICE [98]
	Contrastive	Self Training	DAC [100]
	Contrastive + Clustering Friendly	Pseudo Labeling	SCAN [95]
	Data Specific	Pseudo Labeling	XDC [74]
	Pseudo Labeling	Pseudo Labeling	POT [127]
Simultaneously	Auto-Encoder	Self-training	DEC [102], DBC [106]
	Auto-Encoder	Relation Matching + Self-training	IDEC [103]
	Contrastive + Auto-Encoder	Self-training	DEPICT [105], DEC-DA [104]
	Data Specific + Auto-Encoder	Self-training	SDCN [128]
	Data Specific + Auto-Encoder	Self-training	scDeepCluster [129]
	Pseudo Labeling	Pseudo Labeling + Relation Matching	CCNN [130]
	Clustering Friendly	Pseudo Labeling	PICA [131]
	Contrastive	Pseudo Labeling + Relation Matching	DCCM [132]
	Contrastive	Mutual Information	IIC [133]
	Contrastive	Contrastive	CC [134], DCDC [135]
	Contrastive	Self-training	SCCL [136]
	Contrastive	Relation Matching	DRC [137]
	Contrastive + Auto-Encoder	Self Training + Relation Matching	IcicleGCN [138]
	Contrastive + Clustering Friendly	Contrastive	GCC [139]

* The term (I2I) and (I2C) following “Relation Matching” refer to “Instance to Instance” and “Instance to Cluster” respectively.

Early multistage deep clustering methods [111, 112] have trained a deep auto-encoder to learn representations, which can be directly packed as input of the K-means method to obtain clustering results. Despite their simplicity, these methods had already surpassed earlier approaches in effectiveness and spurred the advancement of deep clustering. Later, deep subspace clustering was proposed to learn an affinity matrix and instance representations first, and then perform clustering by spectral clustering on the affinity matrix [55, 56, 58] or the K-means on the instance representations [57]. Thanks to the contribution of scikit-learn [140] and many other open-source machine learning libraries, clustering algorithms have been applied to many fields and data types with a limited cost of programming. For example, in the scenario of textual/video/graph data clustering, relation (similarity) matching was used in [113, 141, 142], K-Means in [143], Spectral Clustering in [76, 114, 144] and Hierarchical Agglomerative Clustering in [145], so as many other clustering algorithms being directly applied. In graph data, graph cut based node clustering like Metis [146], Graclus [147] and **Balance Normalized Cut (BNC)** [148] were used in graph clustering applications [149–151].

The most recent multistage methods have explicitly incorporated the clustering prior into the representation learning, then conducted clustering on the target friendly representations. For example, IDFD [51] learn representations with two aims: learning similarities between instances and reducing correlations within features. With the above explicit purposes, a naive K-means on the learned representations can also achieve competitive clustering results over many existing deep clustering methods.

Summary. Multistage methods enjoy the property of fast deployment, programming friendly, and intuitive understanding, and thus can be easily employed for applications with various data types [152–159]. However, such a simple combination of deep representation learning and shallow clustering has the following weaknesses:

- (1) The majority of representation learning approaches are not specifically tailored for clustering tasks, lacking the necessary discriminative capability for such tasks.
- (2) The clustering results reveal fundamental connections between the data instances, while they are not used effectively to inform the learning of discriminative representation.

To conclude, such a straightforward cascade connection will cut off the information flow and interactions between representation learning and clustering, thus the limitations of both sides will influence the final performance together.

5.2 Iterative Deep Clustering

To address the shortcomings of the previously described multistage deep clustering, iterative deep clustering improves the process by utilizing the advantages of robust representations to support clustering, while simultaneously the results of clustering inform the representation learning process. In summary, the typical iterative deep clustering framework is iteratively updated through two main phases: (1) deriving clustering outcomes from current representations and (2) refining the representations based on the latest clustering outcomes. As the representation module serves only to feed the clustering module in iterative deep clustering, this subsection categorizes the existing iterative deep clustering approaches based on the feedback from the clustering module.

5.2.1 Iterative Deep Clustering with Individual Supervision. The individual supervision in iterative deep clustering depends on the pseudo labels generated by the clustering module, which can be used to train the representation learning module in a supervised manner.

In early works [49, 125], the cluster centroids and assignments are updated in a K-means way. S²ConvSCN [53] and PSSC [126] combine subspace clustering and pseudo labeling, which obtain pseudo labels by spectral clustering or partitioning the pseudo similarity graph. Later, many works

[95, 98, 99] tend to utilize neural networks for both representation learning and clustering, where these two parts are combined together as one neural network. The clustering module is usually a **multilayer perceptron (MLP)** that produces soft clustering assignments. In this way, the hard pseudo labels can guide both clustering and representation learning through gradient backpropagation with proper constraints. The representative method is DeepCluster [99], which alternates between K-means clustering and updating the backbone along with the classifier by minimizing the gap between predicted clustering assignments and pseudo labels. In fact, DeepCluster has already been applied as a mature clustering algorithm in video clustering [74].

Later, SCAN [95] follows a pretraining-with-finetuning framework. The clustering results are fine-tuned with self-labeling, which selects the highly confident instances by threshold the soft assignment probability, and updates the whole network by minimizing the cross-entropy loss on the selected instances. SPICE [98] is another representative iterative deep clustering method, where the classification model is first trained under the guidance of pseudo labels and then retrained by the semi-supervised training on the set of reliably labeled instances.

5.2.2 Iterative Deep Clustering with Relational Supervision. The relational supervision in iterative deep clustering refers to the relationship based on the pseudo labels, which provides pairwise guidance to the representation learning module. More specifically, the relationship is usually modeled by whether two instances have same discrete pseudo labels [7, 100] and the model is trained as a binary classification task. Another popular branch of methods [54, 101] models the relationship by the similarity of cluster assignment probabilities, which trains representation learning as a regression task.

Summary. Iterative deep clustering methods [123, 124, 127, 160–162] benefit from mutual promotion between representation learning and clustering. However, they also suffer from the error propagation in the iterative process. More specifically, inaccurate clustering results can lead to chaotic representations in which performance is limited by self-labeling effectiveness. Furthermore, this will in turn affect clustering results, especially in the early stage of training. Therefore, existing iterative clustering methods rely heavily on the pretraining of the representation module. Despite these challenges, iterative deep clustering methods enjoy rapid improvements in both representation learning and clustering capabilities, showing considerable potential in practical applications.

5.3 Generative Deep Clustering

While the previously discussed methods have established a connection between the representation learning and clustering modules through gradient propagation, this connection relies solely on the transfer of gradients between the modules. In an effort to enhance the natural collaborative capabilities of these modules, leveraging the broad utility of deep generative networks in both areas, approaches based on generative deep clustering have been developed and extensively explored.

Specifically, this group of methods makes hypotheses about the latent cluster structure and then infers the clustering assignment by estimation of data density. The most representative model is the **Gaussian mixture model (GMM)** [163], which assumes that the data points are generated from a mixture of Gaussians. Specifically, suppose that there are K clusters, and an observed sample $\mathbf{x}$ is generated from the following process:

- (1) Choose a cluster: $c \sim \text{Mult}(\pi)$
- (2) Draw a sample: $\mathbf{x}|c \sim \mathcal{N}(\mu_c, \sigma_c^2 \mathbf{I})$

where π denotes the prior probability for clustering; $\text{Mult}(\pi)$ is the multinomial distribution with the parameter π ; μ_c and σ_c are the mean and variance parameters of the Gaussian distribution

corresponding to the cluster c . The well-known expectation maximization algorithm can be used to learn optimal parameters and clustering assignments.

Although GMM has gained successful applications, its shallow structure is usually insufficient to capture the nonlinear patterns of the data, adversely affecting its performance on complex data (e.g., images, texts, graphs, etc.). To address this problem, deep generative models have been proposed to combine the generative model with powerful deep neural networks, which have enough capacity to model complex and nonlinear data. This kind of method can be classified into two types: methods based on VAE and methods based on **Generative Adversarial Networks (GAN)**.

5.3.1 Deep Generative Clustering Based on Variational Auto-Encoder. For clustering of high-dimensional and complex data, one promising solution is to directly stack GMM with a deep neural network — GMM generates a latent vector z , and the deep neural network further transforms the latent vector z into the complex data instance x . In this way, the stacked model can enjoy the merits of the latent cluster structure and meanwhile has sufficient capacity to model the complex data. For example, the representative models, VaDE [116] and GMVAE [115], assume the following generative process for each instance:

- (1) Choose a cluster: $c \sim \text{Mult}(\pi)$
- (2) Draw a latent vector: $z|c \sim \mathcal{N}(\mu_z(c; \beta), \sigma_z^2((c; \beta))\mathbf{I})$
- (3) Draw a sample: $x|z \sim \mathcal{N}(\mu_x(z; \theta), \sigma_x^2(z; \theta)\mathbf{I})$

where $\mu_z(\cdot; \beta)$, $\sigma_z^2(\cdot; \beta)$, $\mu_x(\cdot; \theta)$ and $\sigma_x^2(\cdot; \theta)$ are given by neural networks with parameters β and θ , which determinize the mean and variance of Gaussian distributions, respectively. Given the above generative process, the optimal parameters and cluster assignment can be obtained by maximizing the likelihood of the given data points as in Equation (19):

$$\log p(\mathbf{x}) = \log \int_z \sum_c p(\mathbf{x}|z)p(z|c)p(c)dz. \quad (19)$$

However, directly optimizing the above likelihood is intractable as it involves integration and complex neural networks. VAE [30] sheds light on how to tackle this problem so that the parameters and posterior can be efficiently estimated via backpropagation. Specifically, the generative model is trained with the following variational inference objective as Equation (20), a.k.a. the ELBO:

$$\mathcal{L}_{\text{ELBO}}(\mathbf{x}) = \mathbb{E}_{q(z, c|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, z, c)}{q(z, c|\mathbf{x})} \right], \quad (20)$$

where $q(z, c|\mathbf{x}; \varphi)$ is the variational posterior to approximate the true posterior, which can be modeled with the recognition networks φ . Monte Carlo [32] and the reparameterization trick [30] can be used to learn the parameters.

More recently, upon VaDE and GMVAE, some improved variants have been proposed. For example, Prasad et al. [164] introduced a data augmentation technique that constrains an input instance (e.g., image) to share a similar clustering distribution with its augmented one; Li et al. [165] employed Monte Carlo objective and the Std Annealing track for optimizing mixture models, which would generate better-separated embeddings than the basic VAE-based methods; Ji et al. [166] proposed to replace decoder in VAE with an improved mutual-information-based objective; Wang et al. [167] proposed to separate the latent embeddings into two parts which capture the particularity and commonality of the clusters, respectively.

5.3.2 Deep Generative Clustering Based on Generative Adversarial Network. Recent years have witnessed great success of the GAN in estimating complex data distribution [122, 168–171]. A

standard GAN contains two components: a generator G that aims at synthesizing “real” samples to fool the discriminator, and a discriminator D tries to discriminate the real data from the generated samples. With the adversary between the two components, the generator could generate samples that have a distribution similar to the data. Inspired by such a great ability, it would be promising to integrate GAN into generative clustering models. Specifically, Ben-Yosef et al. [119] proposed to stack a GMM with a GAN, where a GMM serves as a prior distribution for generating data instances. Formally, they optimized the following objective function as Equation (21):

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_X(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_Z(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (21)$$

where $p_X(\mathbf{x})$ denotes the training data distribution; $p_Z(\mathbf{z})$ is a prior distribution of G and defined as mixture of Gaussians as Equation (22):

$$p_Z(\mathbf{z}) = \sum_{k=1}^K \pi_k * \mathcal{N}(\mu_c, \sigma_c^2 \mathbf{I}). \quad (22)$$

By equipping GAN with such a multi-modal probability distribution, the model could provide a better fit to the complex data distribution especially when the data includes many different clusters.

There are also some improved variants. For example, Yu et al. [120] proposed to directly replace the Gaussian distribution of the GMM with a GAN and developed a ϵ -expectation-maximization learning algorithm to forbid early convergence issues; Ntelemis et al. [172] proposed to employ Sobel operations prior to the discriminator of the GAN; Mukherjee et al. [118] proposed to sample the latent vector $\mathbf{z}$ from a mixture of one-hot encoded variables and continuous latent variables. An inverse network with clustering-specific loss is introduced to make the model more friendly to the clustering task. Analogously, an inverse network is introduced in [121, 173] for the feature-level (i.e., latent vector) adversary.

Summary. Although deep generative clustering models can generate samples while completing clustering, they also have some weaknesses: (1) Training a generative model usually involves Monte Carlo sampling, which may incur training unstable and high computational complexity; (2) The mainstream generative models are based on VAE and GAN, and inevitably inherit the same disadvantages of them. VAE-based models usually require prior assumptions on the data distributions, which may not be held in real cases; although GAN-based algorithms are more flexible and diverse, they usually suffer from mode collapse and slow convergence, especially for the data with multiple clusters.

5.4 Simultaneous Deep Clustering

In contrast to generative deep clustering methods that strengthen the connection between representation learning and clustering through probability distributions, a more effective technique to leverage the advantages of deep learning is to optimize both modules simultaneously under a single optimization goal. As a result, simultaneous deep clustering has developed rapidly and has achieved significant success in recent times. Although most iterative deep clustering methods also optimize both two modules with a single objective, the two modules are optimized explicitly iteratively and cannot be updated simultaneously. In this subsection, we introduce the representative architectures of simultaneously deep clustering.

5.4.1 Auto-Encoder with Self-Training. Auto-encoder is a powerful tool to learn data representations in an unsupervised way, and has been utilized since the first attempts of simultaneously deep clustering [112].

The representative method is DEC [102] which combines the auto-encoder with the self-training strategy. This simple but effective strategy has deeply influenced the follow-up works.

The auto-encoder is pre-trained, and only the encoder is utilized as the initialization of the representation learning module. The self-training strategy mentioned in Section 4.3 is then introduced to optimize clustering and representation learning simultaneously.

Based on the vanilla DEC method, many variants and improvements are proposed. To preserve the local structure of each instance, IDEC [103] further integrates the reconstruction loss into the autoencoder. The general formulation of auto-encoder with self-training can be summarized as Equation (23):

$$\mathcal{L}_{AEST} = \mathcal{L}_{AE} + \mathcal{L}_{ST}, \quad (23)$$

where $\mathcal{L}_{AE}$ is the loss of auto-encoder and $\mathcal{L}_{ST}$ is the loss of clustering oriented self-training, e.g., the neighborhood constraint in DEC. To improve the capability of auto-encoder, some efforts are made to adapt different data types. In [104–106], the linear layer of auto-encoder is replaced with fully convolutional layers so that the image feature can be captured well. In CCNN [130], the clustering convolutional neural network is proposed as a new backbone to extract the representations that are friendly to the clustering task. In DEPICT [105], an additional noisy encoder is introduced, and the robustness of the auto-encoder is improved by minimizing the reconstruction error of every layer between the noisy decoder and the clean encoder.

Although the self-training strategy has achieved success, later work also makes attempts to solve a specific problem. To increase the robustness of the clustering, self-training is applied between two branches: clean and augmented [104] (noisy [105]). Specifically, the target distribution $\mathbb{P}$ is computed using the clean branch to guide the soft assignments $\mathbb{Q}$ of the augmented or noisy branch. The self-training strategy can be combined with subspace clustering. CSC [107] introduces the assumption named invariance of distribution, i.e., the target distribution $\mathbb{P}$ should be invariant to different distance metrics in subspace space. Therefore, two metrics (Euclidean and Cosine distance) are used to calculate the target distributions $\mathbb{P}_E$ and $\mathbb{P}_C$, with the KL divergence between them minimized.

According to the analysis aforementioned, self-training is similar to the K-means clustering and suffers from the unbalance problem [174] between different clusters. To solve the problem of unbalanced data and out-of-distribution samples, StatDEC [9] improves the target distribution by adding the normalized instance frequency of the clusters. In this way, the model can preserve discrimination of small groups and form a local clustering boundary which is insensitive to unbalanced clusters.

5.4.2 Mutual Information Maximization Based Clustering. As illustrated in Section 3.3 the mutual information maximization has been successfully applied in both representation and clustering. The unified form of mutual information in both modules has provided convenience for understanding and implementation.

The representative mutual information maximization based clustering method is DCCM [132]. For each data instance, the mutual information between the deep and shallow layer representations is maximized so that the consistency of the representation can be guaranteed. This consistency is further extended to the cluster assignment space by encouraging instances with the same pseudo-labels to share similar representations.

Many later works can be regarded as variants of this method. In VCAMI [175] and IIC [133], the **augmented mutual information (AMI)** is introduced to improve robustness. Such an augmentation invariant has inspired the later contrastive deep clustering methods, which will be introduced in the next subsection. In ImC-SWAV [176], the mutual information between the integrated discrete representation and a discrete probability distribution is maximized, which improves the vanilla SWAV [177] method.

5.4.3 Contrastive Deep Clustering. Similar to mutual information maximization based deep clustering, contrastive learning has also been successfully applied in both the representation learning module and the clustering module. The main idea of contrastive learning is pulling similar instances closer while pushing different instances away, which is in the spirit of clustering that instances from the same cluster should be close while instances from different clusters should be apart.

The representative contrastive deep clustering method is CC [134]. The basic idea is to treat each cluster as the data instance in the low-dimensional space. The instance discrimination task in contrastive representation learning can be migrated to the clustering task by discriminating different clusters, which is the fundamental requirement of clustering. Furthermore, the advantage of augmentation invariant and local robustness can be preserved in the clustering task.

Taking CC as the fundamental architecture, many contrastive deep clustering can be viewed as variants of it. PICA [131] can be viewed as the degeneration of CC without augmentation, it directly separates different clusters by minimizing the cosine similarity between the cluster assignment statistic vectors. In DCCM [132], the augmentation is introduced to guarantee the local robustness of the learned representation. DRC [137] has the same contrastive learning as CC where the cluster representation is called assignment probability. The difference lies in the regularization of the cluster, which is inspired by group lasso [178]. In CRLC [179], contrastive learning is performed between the cluster assignments of two augmented versions of the same instance, rather than the cluster representations. Also, the dot product in contrastive learning is replaced by the log-dot product, which is more suitable for probabilistic contrastive learning. SCCL [136] extends this approach with textual data augmentation, which proves that this contrastive learning based self-training framework is universally applicable.

The later works further adopt the contrastive clustering in the semantic space. In SCL [180], the negative samples are limited by different pseudo labels, so that instances from different clusters can be further distinguished. MiCE [181] follows a divide-and-conquer strategy in which the gating function divides the whole dataset into clusters and the experts in each cluster aim at discriminating the instances in the cluster. However, compared with InfoNCE which implicitly models alignment and uniformity implicitly [45], MiCE models these two properties in a more explicit way. Recently, TCC [182] further improved efficiency with the reparametrization trick and explicitly improved the discriminability of the cluster.

Some other methods attempt to overcome the problems of vanilla contrastive learning. In GCC [139], the positive pair and negative pair are selected by the KNN graph constructed on the instance representation. This may be related to overcoming the “false negative” problem in contrastive learning. In NCC [183], the contrastive learning module is replaced from SimCLR to BYOL [184], so that the overreliance on negative samples can be solved.

5.4.4 Hybrid Simultaneous Deep Clustering. The aforementioned simultaneous deep clustering methods have remarkable characteristics and advantages, some other works are hybrids of the above techniques. SCCL [136] and Sundareswaran et al. [185] combine contrastive representation learning and self-training. DDC [186] and RCC [187] combine relation matching clustering and pseudo labeling to boost the clustering performance. DCC [188] combines autoencoder-based representation and relation matching clustering. The auto-encoder based representation learning and spectral clustering are combined in [117] by incorporating the augmentations from the contrastive learning.

Summary. The simultaneous deep clustering [128, 129, 135, 138, 189–198] has attracted the most attention for its unified optimization. Intuitively, the learned representation is clustering oriented, and the clustering is conducted on the discriminative space. However, it may arise from

an undesired prejudice of the optimization focus between the representation learning module and the clustering module, which can only be mitigated by manually setting the balanced parameter for now. Also, the model is easy to sink into degenerate solutions where all instances are assigned to a single cluster.

6 Datasets and Evaluation Metrics

In this section, we introduce benchmark datasets and evaluation metrics that are widely used in existing deep clustering methods.

6.1 Datasets

6.1.1 Image Datasets. Image is the most commonly used data type in real-world deep clustering. The early attempts of deep clustering are applied to image datasets including COIL-20,³ CMU PIE,⁴ Yale-B,⁵ MNIST,⁶ CIFAR⁷ and STL-10.⁸ Recently, efforts has been paid to perform clustering on large volume vision datasets (e.g., ImageNet). Although existing methods have achieved promising performance on ImageNet-10 and ImageNet-Dogs dataset, clustering on Tiny-ImageNet (200 clusters) or full-size ImageNet is still challenging.

6.1.2 Textual Datasets. The widely used textual datasets in early applications of textual data clustering include Reuters-21578⁹ and 20 Newsgroups¹⁰ datasets, which have already been vectorized and little feature engineering is needed. Currently, raw textual datasets including IMDB,¹¹ stackOverflow,¹² and more in nlp-datasets github repository¹³ is still challenging for deep textual clustering.

6.1.3 Video Datasets. The ultimate task of video clustering varies from action classification [76] to video anomaly detection [153]. Kinetics-400 and Kinetics-600¹⁴ are two of the most famous video datasets. The others include UCF-101 dataset¹⁵ and HMDB-51 dataset.¹⁶

6.1.4 Graph Datasets. Commonly used graph datasets for node clustering can be referred from the following papers [149, 150, 199] and Stanford Network Analysis Project.¹⁷ And there are also graph-level classification datasets like PROTINS [90] and MUTAG [200], which can be used to perform and evaluate graph-level clustering.

6.2 Evaluation Metrics

Evaluation metrics aim at evaluating the validity of methods. In the field of Deep Clustering, three standard clustering performance metrics are widely used: Accuracy(ACC), **Normalized Mutual Information (NMI)** and **Adjusted Rand Index (ARI)**.

³<http://www.cs.columbia.edu/CAVE/software/softlib/coil-20.php>

⁴<http://www.cs.cmu.edu/afs/cs/project/PIE/MultiPie/Multi-Pie/Home.html>

⁵<http://vision.ucsd.edu/~leekc/ExtYaleDatabase/Yale%20Face%20Database.htm>

⁶<http://yann.lecun.com/exdb/mnist/index.html>

⁷<http://www.cs.toronto.edu/~kriz/cifar.html>

⁸<https://cs.stanford.edu/~acoates/stl10/>

⁹<https://archive.ics.uci.edu/ml/datasets/Reuters-21578+Text+Categorization+Collection>

¹⁰<http://qwone.com/~jason/20Newsgroups/>

¹¹<http://ai.stanford.edu/~amaas/data/sentiment/>

¹²<https://github.com/jacoxu/StackOverflow>

¹³<https://github.com/niderhoff/nlp-datasets>

¹⁴<https://www.deepmind.com/open-source/kinetics>

¹⁵<https://www.crcv.ucf.edu/research/data-sets/ucf101/>

¹⁶<https://serre-lab.clps.brown.edu/resource/hmdb-a-large-human-motion-database/#dataset>

¹⁷<http://snap.stanford.edu/>

6.2.1 Accuracy. ACC indicates the average correct classification rate of clustering samples. Given the ground truth labels $Y = \{y_i | 1 \leq i \leq N\}$ and the predicted hard assignments $\tilde{Y} = \{\tilde{y}_i | 1 \leq i \leq N\}$, ACC can be computed as Equation (24):

$$ACC(\tilde{Y}, Y) = \max_g \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{y_i = g(\tilde{y}_i)\}, \quad (24)$$

where g is the set of all possible one-to-one mappings between the predicted labels and ground truth labels. The optimal mapping can be efficiently obtained by the Hungarian algorithm [201].

6.2.2 Normalized Mutual Information. NMI quantifies the mutual information between the predicted labels and ground truth labels into $[0, 1]$ as Equation (25):

$$NMI(\tilde{Y}, Y) = \frac{\mathcal{I}(\tilde{Y}; Y)}{\frac{1}{2} [H(\tilde{Y}) + H(Y)]}, \quad (25)$$

where $H(Y)$ is the entropy of Y and $\mathcal{I}(\tilde{Y}; Y)$ is the mutual information between $\tilde{Y}$ and Y .

6.2.3 Adjusted Rand Index. ARI comes from the **Rand Index (RI)**, which regards the clustering result as a series of pair-wise decisions and measures it according to the rate of correct decisions as Equation (26):

$$RI = \frac{TP + TN}{C_N^2}, \quad (26)$$

where TP and TN denote the number of true positive pairs and true negative pairs, C_N^2 is the number of possible sample pairs. However, the RI value of two random partitions is not a constant approaching 0, thus ARI was introduced as Equation (27):

$$ARI = \frac{RI - \mathbb{E}(RI)}{\max(RI) - \mathbb{E}(RI)}. \quad (27)$$

Both ACC and $NMI \in [0, 1]$ while $ARI \in [-1, 1]$, in which higher values indicate better performance.

7 Applications

Despite the success of deep clustering in mining global patterns among instances, it has also benefited various downstream tasks. In this section, we discuss some typical applications of deep clustering.

7.1 Community Detection

Community Detection [6, 202, 203] aim at partitioning the graph network into several subgraphs mainly according to connection density, which can be treated as a node-level graph clustering task. Early works are mainly based on modularity measurement [204, 205], maximum flows [206], graph cut [207] and its extension, spectral methods [208]. With the development of GNNs [209, 210], nodes are represented as individual instances in the low-dimensional space. As a result, the border between modern community detection [5] and graph clustering [211–213] is gradually blurring, and GNN-based graph clustering [214, 215] have already been applied to many applications. However, unlike early community detection, which focuses on the network topology, graph clustering usually incorporates the node attributes and other side information. How to release the power of GNNs while preserving the topology characteristic is still under study.

7.2 Anomaly Detection

Anomaly detection (a.k.a. Outlier Detection, Novelty Detection) is a technique for identifying abnormal instances or patterns among data. Early before deep clustering, density-based clustering methods [216–218] have specifically mentioned and addressed the problem of noise during clustering, which has enlightened a group of density-based anomaly detection methods [219, 220]. The later anomaly detection methods [153, 221–223] have utilized the clustering results and identified anomaly instances as those far from the cluster centroids or the border of each cluster. Currently, deep clustering has shown great potential in forming a better clustering space for anomaly detection. Instead of performing anomaly detection after deep clustering, recent efforts have been put into conducting them in a unified framework: identify and remove anomaly instances to reduce the impact on clustering [224], and anomaly detection can be further improved with better clustering results.

7.3 Segmentation and Object Detection

Image Segmentation is one of the most important approaches to simulating human understanding of images that aim at dividing pixels into disjoint regions. Generally speaking, image segmentation is performing pixel classification in a supervised manner and pixel clustering in an unsupervised manner [225, 226]. Currently, deep clustering has been successfully applied in segmentation using clustered regions to generate a Scene Graph [227]. Yi et al. [228] surveyed graph cut based image segmentation, where graph cut is one of the most fundamental solutions to perform clustering (Spectral Clustering). 3D clustering can be a solution to 3D Object Detection, like in [229], 3D points were clustered to represent an object with geometric consistency. But such clustering based segmentation and object detection have no guarantee for small regions and objects, where the expected clustering result is highly unbalanced. The global positional information of a pixel may be ignored when clustering is performed.

7.4 Medical Applications

The convolutional neural network has successfully promoted the development of medical image processing in a supervised manner. However, the manual dataset labeling process is often labor intensive and requires expert medical knowledge [123], which is hard to realize in real-world scenarios. Recently, deep clustering has been introduced to automatically categorize large-scale medical images [123]. Mittal et al. [160] introduce medical image clustering analysis for a faster diagnosis of COVID-19. In the field of biological sciences, **single-cell RNA sequencing (scRNA-seq)** [230] provides a cell gene matrix in output for the analysis of cell population and behavior and even new cell discovery. For this purpose, ScDeepCluster [129] and ItClust [189] develop their models based on DEC [102] to cluster scRNA-seq data and MARS [231] that combine transfer learning and clustering to discover novel cell types. More applications can be found in the gene data clustering area [232].

Beyond the successful applications of deep clustering previously mentioned, clustering holds significant promise in various other fields such as financial analysis [233–235], trajectory analysis [236, 237], and understanding social media [238–240]. Although many of these methods have not yet adopted deep learning techniques, given the growing volumes and complexity of data, deep clustering is expected to significantly influence these areas.

8 Future Directions

In this section, we conduct some future directions of deep clustering based on the above cornerstone, taxonomy, and real-world applications.

8.1 Initialization of Deep Clustering Module

The initialization of deep neural networks usually plays an important role in the efficiency and stability of training [241]. This is more critical in deep clustering, where both the representation learning module and the clustering module are modeled by deep neural networks. Recently, model pre-training [242] has been a popular network initialization technique that has also been introduced to deep clustering [95]. However, the pre-training based initialization is appropriate for the representation learning module but has not been well studied for the clustering module. Although there have been some initialization schemes on shallow clustering [243], the initialization for the cluster module with deep neural networks is still under investigation.

8.2 Overlapping Deep Clustering

The deep clustering methods discussed in this article largely focus on partitioning clustering where each instance belongs to a single cluster. Meanwhile, in the real-world scenario, each instance may belong to multiple clusters, e.g., users in a social network [244] may belong to several communities, and video/audio on social media may have several tags [245]. Among the deep clustering methods discussed in this article, if the clustering constraints are conducted on the cluster assignment probabilistic matrix, they can be directly adapted to the overlapping clustering setting. However, if the training relies on the pseudo hard label of data instances, they may fail in the overlapping clustering setting. Although multi-label classification has been widely studied in the literature [246, 247], how to adapt to unsupervised clustering is still an open research question.

8.3 Boosting Representation with Deep Clustering

Throughout this article, we can find that a good representation is essential for clustering. Although clustering-friendly representation learning has been studied in the literature, they are designed for the specific shallow clustering method. In contrast, the clustering structure denotes the high-order pattern of the dataset, which should be preserved in the comprehensive representation [248]. The deep clustering methods discussed in this article focus on how to incorporate representation learning to boost the clustering, meanwhile, how to in turn boost the representation learning by the clustering remains to be studied.

8.4 Deep Clustering Explanation

As an unsupervised task, the clustering process usually lacks human priors such as label semantics, number of clusters [249], which makes the clustering results difficult to explain or understand. Some methods [250] have already combined the tags provided by the users to increase the explanation of the clustering results. However, it relies on accurate human tagging, which may not be realized in practice. With the development of causal inference in deep learning, the explanation of clustering among instances will hopefully be improved. Both the research and industry community are expecting a generalized explanation framework for clustering, especially on high-dimensional data. To conclude, how to utilize the casual inference techniques in clustering is of great importance and deserves more attention.

8.5 Transfer Learning with Deep Clustering

Transfer learning [251] aims to bridge the gap between training and test datasets with a distribution shift. The general idea is to transfer the knowledge from the known data to the unknown test data. Recently, deep clustering has played an increasingly important role in unsupervised transfer learning [3, 4, 252–254], where the target domain is unsupervised. For example, ItClust [189] and MARS [231] have achieved success in scRNA-seq clustering (Section 7.4), AD-Cluster [255] that

use clustering has improved domain adaptive person re-identification. In contrast, unsupervised transfer learning methods can also benefit deep clustering. Taking UCDS [256] as an example, the unsupervised domain adaptation is used to perform clustering among variant domains. The distribution shift is one of the key factors that affect the performance of machine learning models, including deep clustering. The performance of cluster analysis can give further understanding of the unsupervised target domain, but how to transfer the clustering result to knowledge and effectively minimize the distribution shift based on clustering results can be further explored.

8.6 Clustering with Anomalies

In Section 7.2, we have discussed the applications of deep clustering in anomaly detection where the instances are well clustered. Concerning the existence of anomaly instances in the dataset, the clustering may also be influenced and mutually restricted, since most existing deep clustering methods have no specific response to the outliers. The classic K-means method is known to be sensitive to outliers, although there have been a few works on overcoming this problem [224], they are designed for shallow clustering methods. To this end, how to improve the robustness of deep clustering for anomaly instances and gradually improve the clustering performance by reducing the detected anomaly instances is still an open research question.

8.7 Degenerate Solution versus Unbalanced Data

The issue of degenerate solutions [99] has emerged as a significant challenge in deep clustering, characterized by the potential assignment of all instances to a single cluster. Numerous deep clustering techniques have implemented additional restrictions to address this issue [95, 99, 112, 131, 133, 134, 257], with the entropy of the cluster size distribution being the most widely adopted strategy. Through maximization of entropy, it is anticipated that instances will be distributed uniformly across each cluster, thereby preventing a collapse into a single cluster. Importantly, the effectiveness of this approach depends on a uniform distribution of actual labels, a condition met by standard datasets such as CIFAR10 and CIFAR100. However, in practical environments, this assumption may be too rigid as many datasets exhibit unbalance or have a long-tailed distribution [258]. The conflicting goals of achieving uniform distribution and handling unbalanced data can significantly undermine the effectiveness of deep clustering. Recently, the focus on unbalanced data clustering [259, 260] has intensified, which is expected to improve the effectiveness of clustering in practical scenarios.

8.8 Efficient Training versus Global Modeling

To improve training efficiency and scalability, most existing deep clustering methods have utilized the mini-batch training strategy, where instances are separated into batches and the model is updated with each batch. This is suitable for the task where the instances are independent on each other, such as classification and regression. However, as deep clustering heavily relies on the complicated relationship among instances, such mini-batch training may lose the ability of global modeling. Although some existing methods have used cluster representations or prototypes [248] to store global information, it is still worth studying how to balance training efficiency and model capability.

9 Conclusion

In this review, we provide a detailed and current summary of the deep clustering research area. We initially outline the foundational elements of deep clustering, specifically the representation learning and clustering modules, along with their typical configurations. We categorize the existing approaches into a taxonomy based on the interaction types between the representation learning

and clustering modules, specifically: multi-stage deep clustering, iterative deep clustering, generative deep clustering, and simultaneous deep clustering. Subsequently, we compile the standard datasets, evaluation metrics, and uses of deep clustering. Finally, we explore prospective future trends in deep clustering that offer promising opportunities.

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